

作业2

将 cache 分为若干组, 每组分为若干块

假设分为 2^a 组, 一个组 2^b 块, 每块大小为 2^k

则 cache 大小为 $2^{(a+b+k)}$ 由题可得 $a+b+k=12$ 或 13

第1轮缺失3次, 第2轮4次, 第3轮4次, $\therefore$ 块大小为 $64=2^6$, $k=6$

将块划分为 $0 \sim 63$, $64 \sim 127$, $128 \sim 191$, $192 \sim 255$, $256 \sim 319$, $320 \sim 383$, $384 \sim 447$, $448 \sim 511$, $512 \sim 575$, $576 \sim 639$, $640 \sim 703$, $704 \sim 767$, $768 \sim 831$, $832 \sim 895$, $896 \sim 959$, $960 \sim 1023$, $1024 \sim 1087$, $1088 \sim 1151$, $1152 \sim 1215$, $1216 \sim 1279$, $1280 \sim 1343$, $1344 \sim 1407$, $1408 \sim 1471$, $1472 \sim 1535$, $1536 \sim 1599$, $1600 \sim 1663$, $1664 \sim 1727$, $1728 \sim 1791$, $1792 \sim 1855$, $1856 \sim 1919$, $1920 \sim 1983$, $1984 \sim 2047$, $2048 \sim 2111$, $2112 \sim 2175$, $2176 \sim 2239$, $2240 \sim 2303$, $2304 \sim 2367$, $2368 \sim 2431$, $2432 \sim 2495$, $2496 \sim 2559$, $2560 \sim 2623$, $2624 \sim 2687$, $2688 \sim 2751$, $2752 \sim 2815$, $2816 \sim 2879$, $2880 \sim 2943$, $2944 \sim 3007$, $3008 \sim 3071$, $3072 \sim 3135$, $3136 \sim 3199$, $3200 \sim 3263$, $3264 \sim 3327$, $3328 \sim 3391$, $3392 \sim 3455$, $3456 \sim 3519$, $3520 \sim 3583$, $3584 \sim 3647$, 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